

Inhoudsopgave

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1 Lorem ipsum dolor sit amet, consectetur.

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Stelling 1.1 (Taylor's Formula over $\mathbb{R}$) Let $E \subseteq \mathbb{R}^n$ be open and let $\overline{B_r(a)} \subseteq E$. Suppose $f \in \mathcal{C}^k(E)$. Then

$$f(x) = \sum_{|\alpha| \leq k} \frac{D^\alpha f(a)}{\alpha!} (x-a)^\alpha + \sum_{|\alpha|=k} h_\alpha(x) (x-a)^\alpha$$

for any $x \in B_r(a)$, where α is a multi-index, and $h_\alpha : B_r(a) \rightarrow \mathbb{R}$ such that $h_\alpha(x) \rightarrow 0$ when $x \rightarrow a$ for all $|\alpha| = k$.

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Opmerking Observe

$$|(x-a)^\alpha| = \left| \prod_{i=1}^n (x_i - a_i)^{\alpha_i} \right| = \prod_{i=1}^n |x_i - a_i|^{\alpha_i} \leq |x-a|^{|\alpha|}$$

Therefore we have that

$$\left| \sum_{|\alpha|=k} h_\alpha(x) (x-a)^\alpha \right| \leq |x-a|^k \sum_{|\alpha|=k} |h_\alpha(x)|.$$

Bewijs (van Taylor's Formula over $\mathbb{R}$) Let $x \in B_r(a)$ and define $[a, x] = \{\varphi(t) \mid t \in [0, 1]\}$, where $\varphi : [-1, 1] \rightarrow \mathbb{R}^n$ is defined by $\varphi(t) = a + t(x - a)$. Now let $g(t) = f(\varphi(t))$ so that $g(0) = f(a)$ and $g(1) = f(x)$. Thus $g \in \mathcal{C}^k([-1, 1])$, meaning

$$g(1) = g(0) + \frac{1}{1!}g'(0) + \dots + \frac{1}{(k-1)!}g^{(k-1)}(0) + \frac{1}{k!}g^{(k)}(c)$$

for some $c \in [0, 1]$. As shown previously, we have

$$f(x) = \sum_{|\alpha| \leq k-1} D^\alpha f(\varphi(t)) \cdot (x - a)^\alpha$$

for $m = 1, \dots, k$ by the Mixed Derivative Theorem. This implies

$$f(x) = \sum_{|\alpha| \leq k-1} \frac{D^\alpha f(a)}{\alpha!} (x - a)^\alpha + \sum_{|\alpha|=k} \frac{D^\alpha f(y)}{\alpha!} (x - a)^\alpha$$

for some $y \in [a, x]$. Now observe $D^\alpha f(y) = D^\alpha f(a) + (D^\alpha f(y) - D^\alpha f(a))$, where $D^\alpha f(y) - D^\alpha f(a) \rightarrow 0$ as $y \rightarrow a$. Lastly define $h_\alpha(x) = \frac{D^\alpha f(y) - D^\alpha f(a)}{\alpha!} \rightarrow 0$ as $x \rightarrow a$, where y is a function of x . □

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Lemma 1.2 (Archimedes) Voor elke $x \in \mathbb{R}$ bestaat er een $n \in \mathbb{N}$ zodat $x < n$.

Bewijs Veronderstel dat er een $x \in \mathbb{R}$ zou bestaan zodat $x \geq n$ voor alle $n \in \mathbb{N}$. Dan zou $\mathbb{N}$ naar boven begrensd zijn en volgens de supremumeigenschap een supremum hebben, zeg $s = \sup \mathbb{N}$. Omdat s de kleinste bovengrens is van $\mathbb{N}$, is $s - 1$ geen bovengrens van $\mathbb{N}$. We kunnen dus een $k \in \mathbb{N}$ vinden zodat $s - 1 < k$, maar dan $s < k + 1$ en omdat $k + 1 \in \mathbb{N}$ is dit strijdig met de aanname dat s een bovengrens van $\mathbb{N}$ is. □